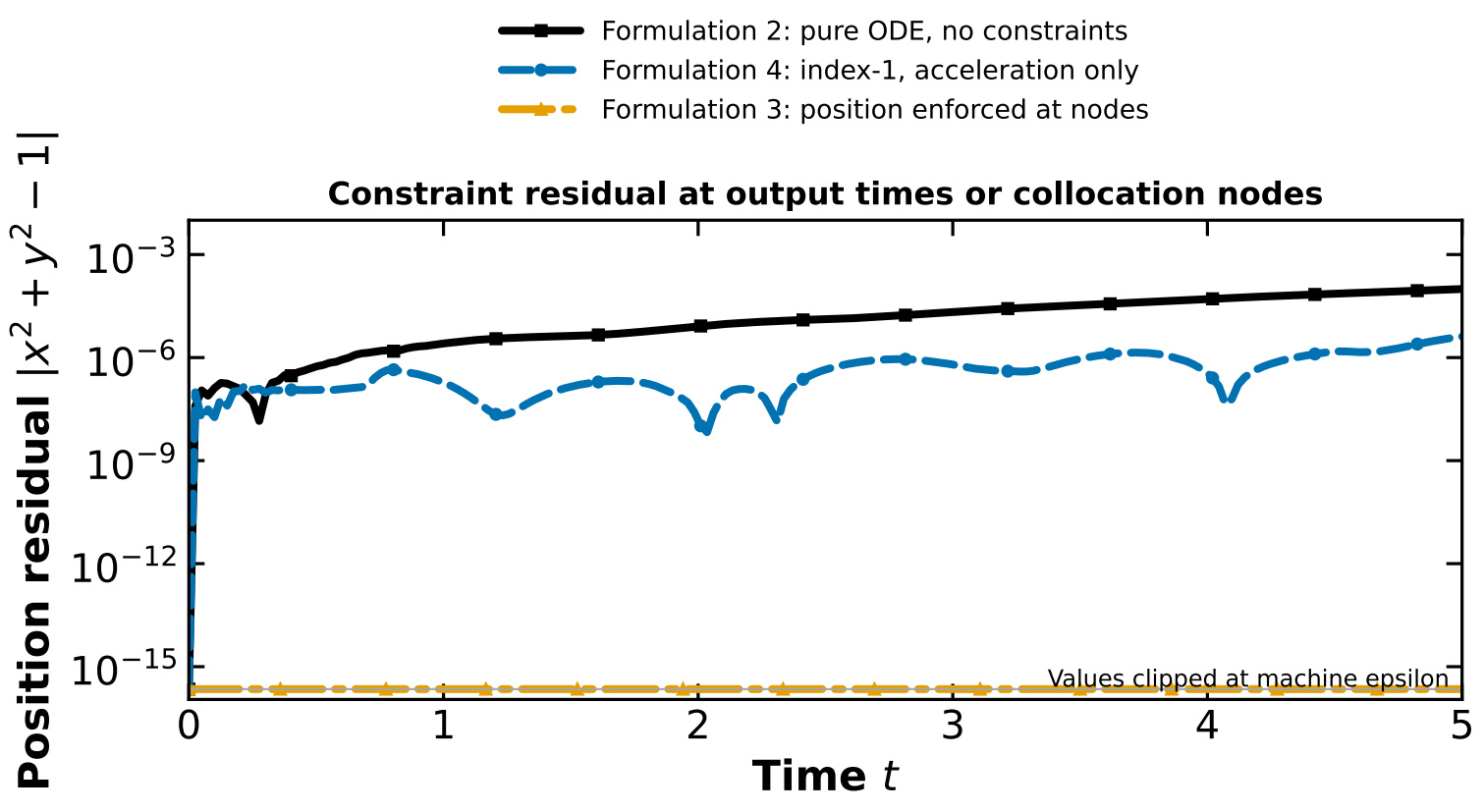
Formulation 2: pure ODE, no constraints
Formulation 4: index-1, acceleration only
Formulation 3: position enforced at nodes
Constraint residual at output times or collocation nodes
Position residual $|x^2 + y^2 - 1|$
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
Values clipped at machine epsilon
0
1
2
3
4
5
Time t